SECURITY



Baremetal Security Architecture

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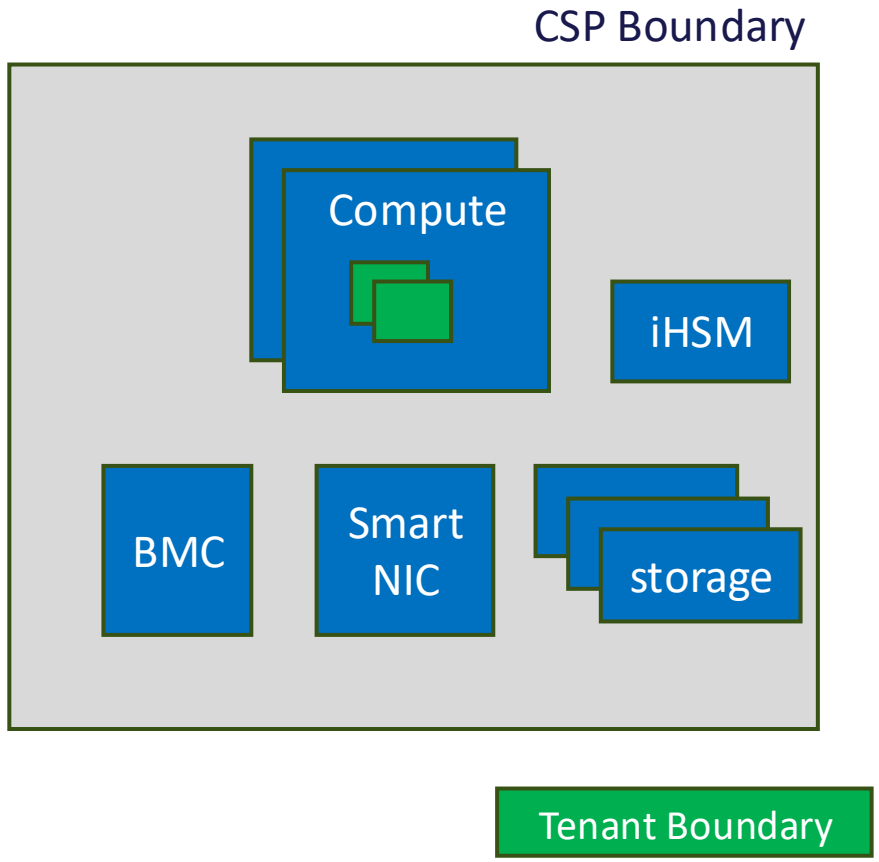


Agenda

- Comparing Tradition General Purpose (GP) Platforms
- Challenges
- Baremetal Tenant control & CSP Control
- A ‘dynamic’ hierarchical model

Traditional General-Purpose (GP) Platforms

- CSP provides VMs
- Tenant boundary is VMs
- CSP owns and manages the platform hardware, firmware, OS etc.
- GP is like a taxi model
  - Customers can only ride the taxi



Bare Metal platforms

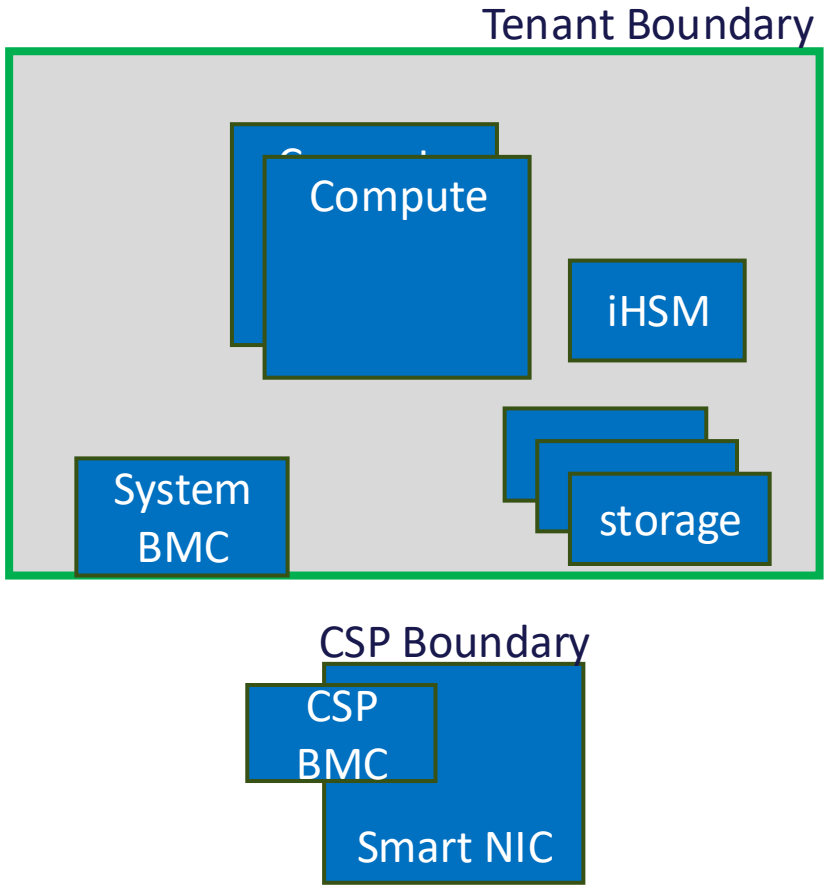
- Baremetal is like a Rental Car model
  - Tenants rent the HW, CSP manages it
- Baremetal tenants want full platform access
- CSPs still need to
  - Manage the hardware & firmware
  - Own backend services and performance throughputs (e.g. CSP fabric)
  - Isolate tenant vs CSP boundary
- CSP are still responsible for the hardware & security

Challenges

- Bare metal tenants want control & manage platform
  - CSPs traditionally owned BMC and thereby controlled the services for security, recoverability & manageability
- Keeping FW up to date
  - How to do it uninterrupted? – Impactless updates
  - How does tenant permit the update? – DoT
  - How to deploy FW without traditional compute access? – OOB updates
- Regaining access to platform when tenant vacates
- What happens if the system is bricked?

Can we have the cake and eat it too?

- A Dynamic Hierarchical Model
  - Normal Mode: CSP BMC provides services to System BMC
  - Recovery Mode: CSP BMC takes up platform control
- Provides CSP / Baremetal tenant isolation
  - System BMC provides tenant control
  - CSP BMC manages CSP components
- System Bricked or being vacated?
  - CSP BMC takes over and puts System BMC under recovery
  - Recovers System BMC to CSP image
  - System BMC recovers all other components
  - Streaming boot solves endurance



Hierarchical Model Demo

- See the demo at the Microsoft booth
- CSP BMC integrated into SmartNIC, provides CSP services
- System BMC (in a DC-SCM) provides tenant services
- No change to platform form factor
- Solution Integrated w/ DC-SCM 1.x (while providing hardware partitioning)

Call for action

- System BMC is implemented in various FF across AI & GP platforms
- Food for thought for Future system designs
  - Recoverability of the whole system
  - Security isolation of CSP vs Baremetal Tenant
  - Modular platform construction

Thank You!

